POWER BI
DAX FOR BEGINNERS

Step-by-step Practice with Real Orders and Customers Data

Learn by Doing

COUNT DISTINCTCOUNT SUM AVERAGE

Calculated Columns Profit CALCULATE IF

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Two practice files: `DAX Customers.xlsx` and `DAX Orders.xlsx`

Clean teaching notes, verified using the uploaded datasets

01 What will you learn?

Start here. Every example uses your own Orders and Customers datasets.

What is DAX?

DAX stands for **Data Analysis Expressions**. It is the formula language used in Power BI for creating calculations, totals, and analysis.

> **In simple words:** DAX helps you answer questions such as: How many orders? How many products? How much was sold? How much profit was earned?

Learning path

1. Import the two Excel files and connect them by **Order ID**.
2. Learn the basic DAX formula structure.
3. Count rows, orders, customers, and unique products.
4. Use SUM, AVERAGE, MIN, and MAX for numeric columns.
5. Create a **new column** to calculate sales, costs, and profit per row.
6. Create **measures** to calculate grand totals using SUM.
7. Compare results by customer, product, and city.
8. Make simple charts and practice questions.

Two important DAX objects

Object	Meaning	Use it for
Calculated column	A value calculated for each row	Profit of an individual order
Measure	A result calculated for the current report filters	Total Profit on a Card or by Customer

Our approach: First create calculated columns and learn `SUM()`. Then see how `SUMX()` can calculate total profit directly, without adding a separate Profit column.

02 Understand the two datasets

Know your columns before writing any DAX formula.

Table 1: Customers (100 rows)

Column	Type	What it means
Name	Text	Customer name; repeated across orders
Order ID	Number/ID	Unique order identifier
Order Date	Date	Date on which the order was placed
Product ID	Number/ID	Product identifier
City	Text	Customer location: city
State	Text	Customer location: state

Table 2: Orders (100 rows)

Column	Type	What it means
Order ID	Number/ID	Unique order identifier
Product ID	Number/ID	Identifier of the ordered product
Product	Text	Product name, such as Dhokla
Product State	Text	State associated with the product
Selling Price	Number	Selling price per unit (assumed)
Product Cost	Number	Cost per unit (assumed)
Quantity	Decimal	Number of units in an order

Dataset facts: 100 order rows, 100 matching customer/order records, **5 distinct names**, and **10 distinct products**. Order IDs match between both tables.

Important: `Order ID` and `Product ID` are identifiers. Even though they are numbers, **adding their values is not a useful business calculation**. Likewise, do not sum dates.

03 Import data and make a relationship

Do these steps in Power BI Desktop.

Step 1: Import both Excel files

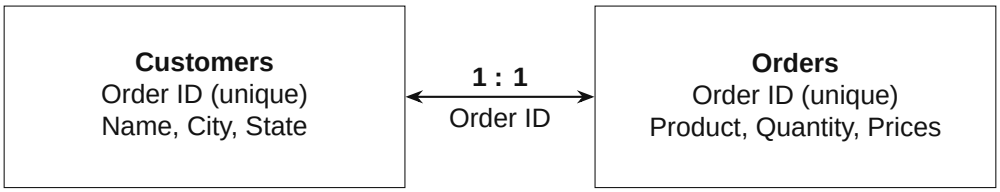
1. Open **Power BI Desktop**.
2. Select **Home** → **Get data** → **Excel workbook**.
3. Select the Customers file, choose its worksheet, and click **Load**.
4. Repeat for the Orders file.
5. Check that your tables are named `Customers` and `Orders`.

Step 2: Check data types

Field	Recommended Power BI data type
Order ID, Product ID	Whole number (identifiers)
Order Date	Date
Selling Price, Product Cost	Decimal number or fixed decimal
Quantity	Decimal number (this file contains 100.5)
Name, City, State, Product	Text

Step 3: Connect the tables

Go to **Model view**. Drag `Customers[Order ID]` onto `Orders[Order ID]`.



Both columns contain 100 unique Order IDs, so this dataset supports a **one-to-one** relationship. Keep it **active**. Use the default filter direction for this 1:1 relationship and check the visual results. **Do not connect by Name:** the same name appears many times, and Orders has no Name column.

> **Example match:** Order ID 80801 belongs to Adwait in Customers and contains Malapua, quantity 21, in Orders.

04 How to write a DAX measure

Start with a very small formula.

Basic syntax

```
Measure Name = DAX_FUNCTION(TableName[ColumnName])
```

Parts of the syntax

- `Measure Name`: the name shown in Power BI Fields/Data pane.
- `=`: starts the calculation.
- `COUNT`, `SUM`, etc.: DAX function names.
- `Orders[Quantity]`: the Quantity column in the Orders table.

Create your first measure

1. Click the `Orders` table in the Data pane.
2. Choose **Modeling** → **New measure**.
3. Type the formula below, then press Enter.
4. Select the measure and add it to a **Card** visual.

```
Total Orders = COUNTROWS(Orders)
```

Expected result: 100 The Orders table has exactly 100 data rows.

A measure can change with report filters

When you put `Customers[Name]` beside `[Total Orders]` in a Table visual, Power BI recalculates the measure for each name because of the relationship. **One rule to remember:** A measure is not simply a column. It calculates an answer in the current *filter context* (for example, a selected customer or product).

05 COUNT, COUNTA, and DISTINCTCOUNT

First, learn how to count data correctly.

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1. Count numeric Order IDs

```
Count of Orders = COUNT(Orders[Order ID])
```

Result: **100**. `COUNT()` counts nonblank numeric values.

2. Count rows directly

```
Total Orders = COUNTROWS(Orders)
```

Result: **100**. This is a clear way to count order rows.

3. Count unique Order IDs

```
Unique Orders = DISTINCTCOUNT(Orders[Order ID])
```

Result: **100**. This counts different Order IDs.

4. Count the nonblank customer names

```
Name Entries = COUNTA(Customers[Name])
```

Result: **100**. `COUNTA()` works for text values.

5. Count different customers and products

```
Unique Names = DISTINCTCOUNT(Customers[Name])

Unique Products = DISTINCTCOUNT(Orders[Product])
```

Results: **5 names** and **10 products**.

> **Remember:** `COUNTROWS()` counts table rows. `COUNT()` counts nonblank numbers. `COUNTA()` counts nonblank values including text. `DISTINCTCOUNT()` counts different values.

06 SUM, AVERAGE, MIN, and MAX

Next, analyze the numeric columns of Orders.

1. Total quantity sold

```
Total Quantity = SUM(Orders[Quantity])
```

Expected result: **2,857.5**

2. Average quantity in an order row

```
Average Quantity = AVERAGE(Orders[Quantity])
```

Expected result: **28.575**

3. Average, minimum, and maximum selling price

```
Average Selling Price = AVERAGE(Orders[Selling Price])

Minimum Selling Price = MIN(Orders[Selling Price])

Maximum Selling Price = MAX(Orders[Selling Price])
```

Measure	Expected result
Average Selling Price	344.25
Minimum Selling Price	20
Maximum Selling Price	1,250

One useful warning

`SUM(Orders[Selling Price])` returns **34,425**; however, this is **not total sales revenue**. It only adds the unit selling prices. Revenue must include each row's `Quantity`, which we will calculate next.

Field-by-field DAX practice

Choose a meaningful calculation for every original column.

Customers table: six original columns

Column	Useful analysis	Actual result
Name	Count distinct names	5
Order ID	Count distinct IDs	100
Order Date	Count distinct dates	90
Product ID	Count distinct IDs	10
City	Count distinct cities	5
State	Count distinct states	5

Orders table: seven original columns

Column	Useful analysis	Actual result
Order ID	Count distinct IDs	100
Product ID	Count distinct IDs	10
Product	Count distinct products	10
Product State	Count distinct states	8
Selling Price	Average unit price	344.25
Product Cost	Average unit cost	128.5
Quantity	Total quantity	2,857.5

Try these additional measures

```
Unique Cities = DISTINCTCOUNT(Customers[City])
Unique Product States = DISTINCTCOUNT(Orders[Product State])
Average Unit Cost = AVERAGE(Orders[Product Cost])
First Order Date = MIN(Customers[Order Date])
Last Order Date = MAX(Customers[Order Date])
```

Dates: **01 Jan 2022** through **31 Mar 2022**, after setting the correct Date data type.

> **Teaching rule:** Count distinct values in identifier and category fields. Use SUM/AVERAGE for business quantities or values. Use MIN/MAX for dates.

07 Analyze orders by customer name

Use a Table visual to see how filters change measures.

Build the visual

1. Click the **Table** visual in Report view.
2. Drag `Customers[Name]` to the visual.
3. Add the `[Total Orders]` measure.
4. Add the `[Total Quantity]` measure.

What you should see

Customer Name	Total Orders	Total Quantity
Adwait	22	612
Chitra	22	741
Gopal	21	478
Madhav	16	567.5
Rishabh	19	459
Grand Total	**100**	**2,857.5**

What did DAX do?

When the row is **Chitra**, Power BI filters the matching order records and computes the total. For that name, 22 orders contain a total quantity of 741.

> **Practice:** Replace `Name` with `City`. You should see the same totals arranged by five cities because each name in this practice file corresponds to one city.

Important: These are 5 different *name values*, not a guaranteed count of 5 real-world individuals. A proper production model should have a unique Customer ID.

08 Create your first calculated column

A new column computes a value for every row.

Step 1: Calculate profit per unit

1. Click the `Orders` table.
2. Choose **Table tools** → **New column**.
3. Enter the following formula and press Enter.

```
Unit Profit = Orders[Selling Price] - Orders[Product Cost]
```

Meaning: From one unit's selling price, subtract one unit's cost.

Check a few order rows

Order ID	Product	Price	Cost	Unit Profit
80801	Malapua	300	250	50
80802	Dhokla	575	90	485
80803	Mysore pak	275	250	25
80804	Modak	250	185	65

Step 2: Make a Profit column

```
Profit = Orders[Unit Profit] * Orders[Quantity]
```

Example for order 80801:
Unit Profit $= 300 - 250 = 50$
Profit $= 50 \times 21 = 1{,}050$

Key distinction: `Unit Profit` is the profit on one item. `Profit` is the profit on all units sold in that order row.

09 Create Sales Amount and Cost Amount

Two more calculated columns make business analysis easy.

Step 1: Sales Amount column

Go to **Table tools** → **New column** in the Orders table.

```
Sales Amount = Orders[Selling Price] * Orders[Quantity]
```

Step 2: Cost Amount column

Again, choose **New column**.

```
Cost Amount = Orders[Product Cost] * Orders[Quantity]
```

Step 3: Profit column (another correct form)

You can make `Profit` directly using the two new columns:

```
Profit = Orders[Sales Amount] - Orders[Cost Amount]
```

Note: This is an *alternative* to `Orders[Unit Profit] * Orders[Quantity]`. Keep **only one** calculated column named `Profit`; do not try to create both with the same name.

Check the first four rows

Order	Product	Sales	Cost	Profit
80801	Malapua	6,300	5,250	1,050
80802	Dhokla	10,925	1,710	9,215
80803	Mysore pak	4,950	4,500	450
80804	Modak	2,500	1,850	650

Business formula: Profit $=$ (Selling Price $-$ Cost) $\times$ Quantity, assuming Selling Price and Product Cost are per-unit values.

10 Calculate totals with SUM and SUMX

Use SUM for prepared columns and SUMX for profit calculated row by row.

Step 1: Make a Total Sales measure

Click **Modeling** → **New measure**.

```
Total Sales = SUM(Orders[Sales Amount])
```

Expected result: **933,890**

Step 2: Make a Total Cost measure

```
Total Cost = SUM(Orders[Cost Amount])
```

Expected result: **338,962.5**

Step 3: Total Profit directly with SUMX

Choose **Modeling** → **New measure**. Use this short form:

```
Total Profit =
SUMX(
    Orders,
    ([Selling Price] - [Product Cost]) * [Quantity]
)
```

Important: The first `Orders` is required by `SUMX(Table, Expression)`. Inside its expression, the column names can be shorter because the function evaluates rows from the Orders table.

Alternative if you already created the Profit column:

```
Total Profit = SUM(Orders[Profit])
```

Use **only one** of these two Total Profit measures. Both give **594,927.5**.

Check the answers

Measure	Result
Total Sales	933,890.0
Total Cost	338,962.5
Total Profit	**594,927.5**

Check: $933{,}890 - 338{,}962.5 = 594{,}927.5$. The SUMX and SUM versions give identical profit totals.